

FULL NAME: _____

BLOCK: _____

Radical Equations — Assessment-Critical: Solve for Variable + Modeling

Lesson 5-4 · Period 3 · Student Packet

OBJECTIVES

Math Objective	Rewrite radical/rational-exponent equations to isolate a specified variable; evaluate such rewritten formulas at given values to produce tables and graphs.
Language Objective	Explain in writing which variable is being isolated and why, using a formula's context (volume, BSA, speed).
Essential Understanding	Solving a formula for a new variable is the SAME algebra as solving a radical equation — isolate the radical/rational-exponent term, raise to the reciprocal power, clean up. Works for both pure-algebra and real-world cases.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Rewrite formulas to isolate a specified variable (the Topic 5 assessment's core skill); evaluate rewritten formulas at given V/x values.
Essential Question	How do we extract ANY variable from a radical/rational-exponent formula, and why does the same procedure work for both algebra and real-world contexts?
Materials	Student packet, pencil, calculator for table evaluation. Practice #44 is the direct assessment rehearsal — treat it like the real test.

LAUNCH — Rewrite a Formula + Table Completion (teacher models Ex 2 + #44)

[BOOK] ISOLATE-A-VARIABLE RULES

1. Identify which variable to isolate (look at the final answer form).
2. Strip away outer operations LAST-IN, FIRST-OUT (addition/subtraction last $\rightarrow$ multiplication/division $\rightarrow$ exponent/radical first to remove).
3. Raise both sides to the RECIPROCAL power to undo a rational exponent ($m/n \rightarrow n/m$).

[BOOK] TABLE-EVALUATION RULES

1. For each input value in the table, substitute into the rewritten formula.
2. Simplify rational exponents step-by-step (use calculator for irrational outputs).
3. Round to specified precision.
4. The table is a point-cloud of the formula's graph — plot the points to see the shape.

EXPLORE — Think-Pair-Share (35 min)

Work with a partner. Practice #10 rehearses Assessment Item 1A (isolate x). Practice #44 (done in Launch) rehearses Assessment Item 2 (complete the table). Plan ~8 min for #39 (BSA application).

Bank-item labels (in order):

1. Try It 2 — Rewrite a formula (partner practice)
2. Practice #10 — Isolate x [STAR] Assessment Item 1A rehearsal
3. Practice #25 — Solve for y , rationalize
4. Practice #26 — Solve for y , rationalize
5. Practice #27 — Solve for y , rationalize
6. Practice #39 — BSA application (modeling)

Try It 2 — Rewrite a formula (partner practice)

The speed, v , of a vehicle in relation to its stopping distance, d , is represented by the equation $v = 3.57\sqrt{d}$. What is the equation for the stopping distance in terms of the vehicle's speed?

Practice #10 — Isolate x [STAR] Assessment Item 1A rehearsal

Rewrite the equation $y = \sqrt{\frac{x-48}{6}}$ to isolate x .

Practice #25 — Solve for y , rationalize

Solve for y . $x = 3\sqrt[3]{15+y}$

Practice #26 — Solve for y , rationalize

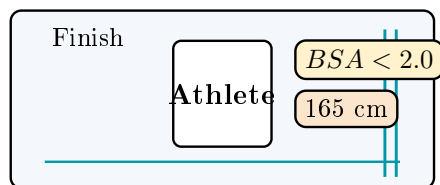
Solve for y . $x = \frac{\sqrt{2y}}{26}$

Practice #27 — Solve for y , rationalize

Solve for y . $x = \frac{\sqrt{y-14.2}}{0.05}$

Practice #39 — BSA application (modeling)

Solve using the formula $BSA = \sqrt{\frac{H \cdot M}{3,600}}$. A sports medicine specialist determines that a hot-weather training strategy is appropriate for this athlete. To the nearest tenth, what can the mass of the athlete be for the training strategy to be appropriate?



SHARE / SUMMARY (5 min)

Sentence frames

Pair share (#10 or #39): “To isolate ____, I first ____, then raise both sides to the power _____. The

rewritten formula is _____. Plugging in _____ = _____ gives _____. In context this means _____.”
Whole class: one pair reads their answer. Class signals thumbs up/sideways.

[CHART] CONCEPT SUMMARY (your reference for Topic 5 assessment)

ISOLATING A VARIABLE FROM A RADICAL/RATIONAL-EXPONENT FORMULA:

- Identify the variable to isolate and the form of the final answer.
- Undo outer operations first (add/sub), then multiply/divide, then remove the exponent/radical.
- To undo $x^{(m/n)}$: raise both sides to the power (n/m) .
- Simplify: distribute, combine like terms, solve for the target variable.

COMPLETING A TABLE FROM A REWRITTEN FORMULA:

- Use your rewritten formula (e.g., $x = \sqrt[3]{4V/2}$) as the rule for each row.
- Substitute each V -value. Use a calculator for irrational outputs. Round as directed.
- Check: does the table’s pattern match the shape you expect (growth/decay/root curve)?
- Assessment Item 2 expects THIS process — from the rewritten formula to the table.

EXIT

To isolate a variable from a rational-exponent formula I _____.

When I evaluate the formula at a specific x , I _____.

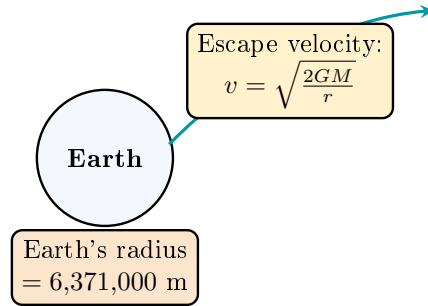
One biggest thing I learned today: _____

OPTIONAL REINFORCEMENT (not collected)

Extra stretch for fast finishers. Try without a calculator first.

Optional #1 (Escape Velocity — performance task)

Performance Task Escape velocity is the velocity at which an object must be traveling to leave a star or planet without falling back to its surface or into orbit. Escape velocity depends on the gravitational constant, $G = 6.67 \times 10^{-11}$, the mass, M in kilograms, and radius, r , of the star or planet.



Part A Rewrite the equation to solve for mass.

Part B Earth has an escape velocity of about 11,200 meters per second. What is Earth's mass in kilograms?